

TOC Assignment

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(031)

Q1. Explain the properties of Regular language and Closure properties of Context free language?

Properties of regular language:

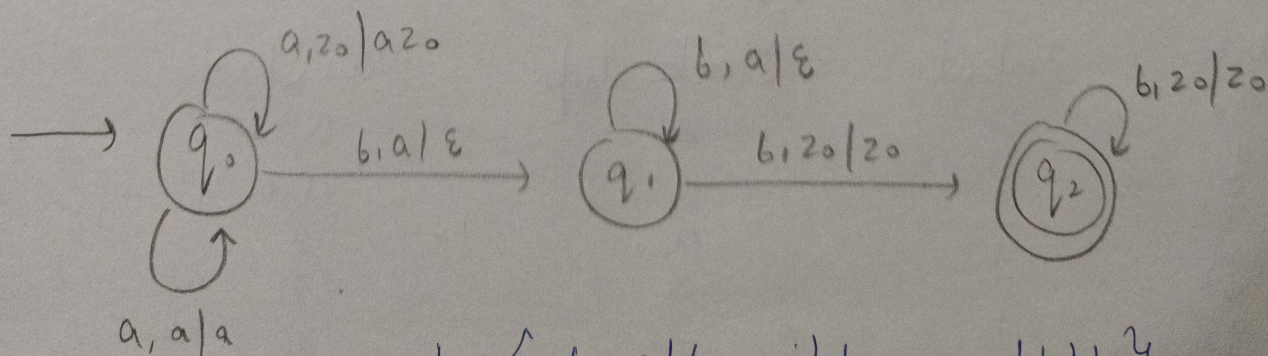
- 1) Union - Let L and M be two regular languages then $L+M$ is also regular language.
- 2) Intersection - Let L and M be two languages of regular expressions R, S . $L \cap M$ is also regular language.
- 3) Concatenation - If L_1 and L_2 are two regular languages, their concatenation $L_1 L_2$ will also be regular.
- 4) Kleene closure - If $L_1 = (a+b)$ then $L_1^* = (a+b)^*$
- 5) Complement - If $L = \{a^n \mid n > 3\}$ then $L' = \{a^n \mid n \leq 3\}$

Properties of Context free language

- 1) Union - Let L_1 and L_2 be two context free languages. Then $L_1 \cup L_2$ is also CFL.
- 2) Concatenation - If L_1 and L_2 are two context free languages then $L_1 L_2$ is also Context free.
 $L_1 = \{a^n b^n\}$ $L_2 = \{c^m d^m\}$ $L_1 L_2 = \{a^n c^n c^m d^m\}$
- 3) Kleene star - If L is Context free, L^* is also context free.
 $L_1 = \{a^n b^n, n \geq 0\}$ $L_1^* = \{a^n b^n\}^*$

Q2. Construct PDA using transition diagram, transition table and write transition for the following languages?

i) $\{a^n b^m \mid n, m \geq 1\}$



$L = \{ab, abb, aabb, aaabbbb\}$

Transitions $S(q_0, a, z_0) = (q_0, az_0)$

$$S(q_0, a, a) = (q_0, a)$$

$$S(q_0, b, a) = (q_1, \epsilon)$$

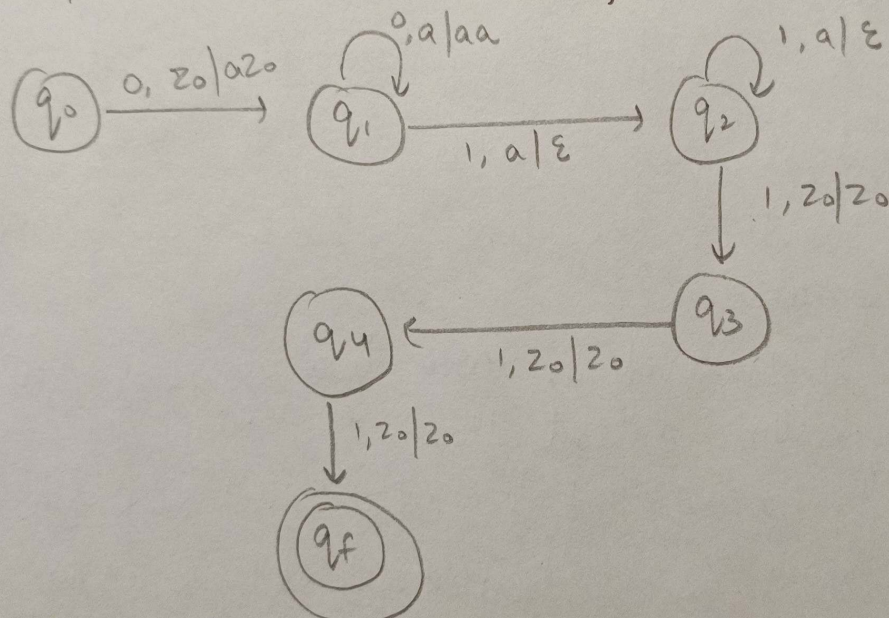
$$S(q_1, b, a) = (q_1, \epsilon)$$

$$S(q_1, b, z_0) = (q_2, z_0)$$

$$S(q_2, b, z_0) = (q_2, z_0)$$

State	Input	Stack Top	New Stack	Next State
q_0	a	z_0	az_0	q_0
q_0	a	a	aa	q_0
q_0	b	a	ϵ	q_1
q_1	b	a	ϵ	q_1
q_1	b	z_0	z_0	q_2
q_2	b	z_0	z_0	q_2

$$2) \{0^n 1^m \mid n \geq 1, m \geq 1, m > n + 2\}$$

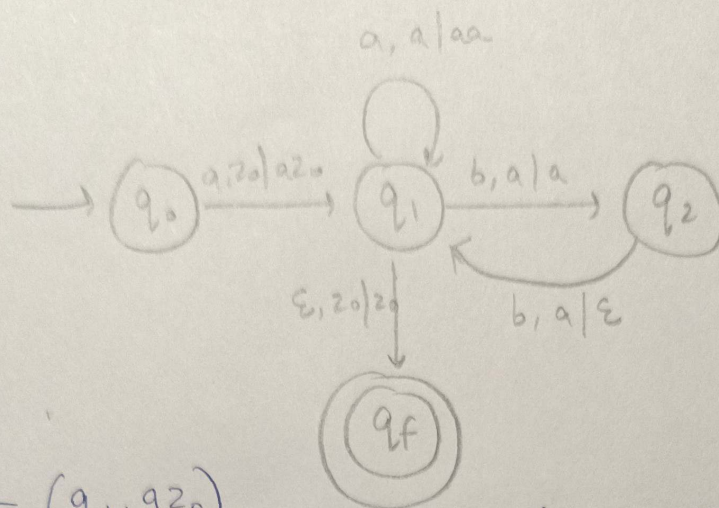


transitions : $S(q_0, 0, z_0) = (q_1, az_0)$
 $S(q_1, 0, a) = (q_1, aa)$
 $S(q_1, 1, a) = (q_2, \epsilon)$
 $S(q_2, 1, a) = (q_2, \epsilon)$

$S(q_2, 1, z_0) = (q_3, z_0)$
 $S(q_3, 1, z_0) = (q_4, z_0)$
 $S(q_4, 1, z_0) = (q_f, z_0)$

State	Input	Stack top	New Stack	Next State
q_0	0	z_0	az_0	q_1
q_1	0	a	aa	q_1
q_1	1	a	ϵ	q_2
q_2	1	a	ϵ	q_2
q_2	1	z_0	z_0	q_3
q_3	1	z_0	z_0	q_4
q_4	1	z_0	z_0	q_f

3.) $\{a^n b^{2n} \mid n \geq 1\}$



transitions :

$$\delta(q_0, a, z_0) = (q_1, az_0)$$

$$\delta(q_1, a, a) = (q_1, aa)$$

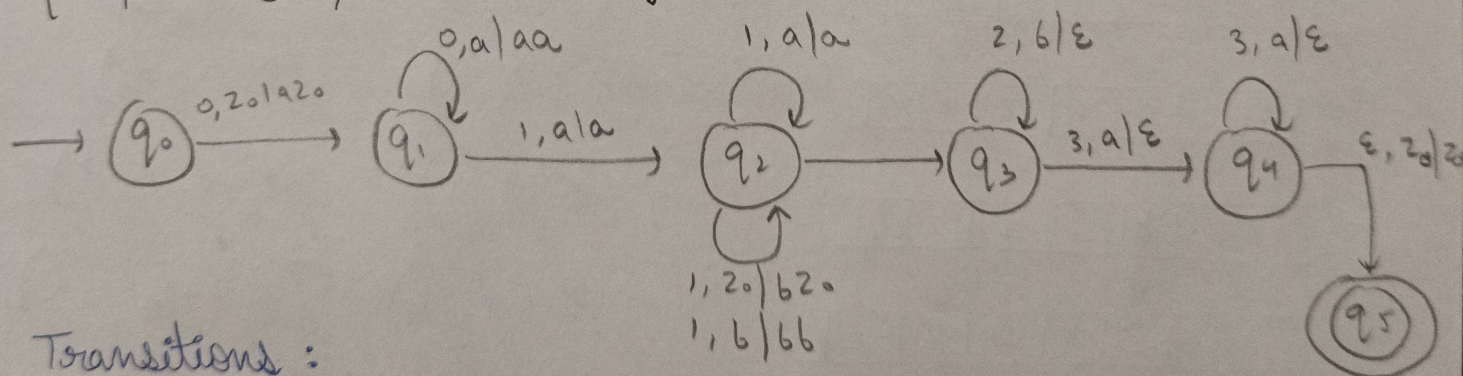
$$\delta(q_1, b, a) = (q_2, a)$$

$$\delta(q_2, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_f, z_0)$$

State	Input	Stack top	New Stack	Next Stage
q_0	a	z_0	az_0	q_1
q_1	a	a	aa	q_1
q_2	b	a	a	q_2
q_3	b	a	a	q_1
q_1	ϵ	z_0	z_0	q_f

4.) $\{0^n 1^m 2^m 3^n \mid n \geq 1, m \geq 1\}$



Transitions :

$$\delta(q_0, 0, z_0) = (q_1, az_0)$$

$$\delta(q_1, 0, a) = (q_1, aa)$$

$$\delta(q_1, 1, a) = (q_2, a)$$

$$\delta(q_2, 1, a) = (q_2, a)$$

$$\delta(q_2, 1, z_0) = (q_2, bz_0)$$

$$\delta(q_2, 1, b) = (q_2, bb)$$

$$\delta(q_2, 2, b) = (q_3, \epsilon)$$

$$\delta(q_3, 2, b) = (q_3, \epsilon)$$

$$\delta(q_3, 3, a) = (q_4, \epsilon)$$

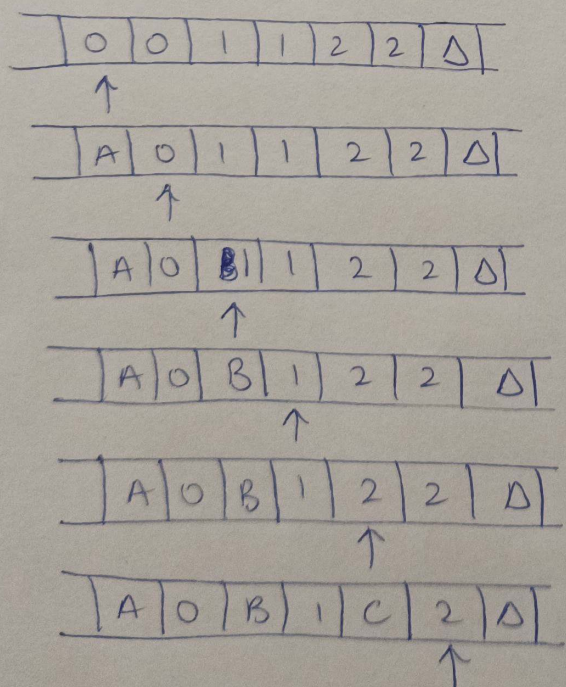
$$\delta(q_4, 3, a) = (q_4, \epsilon)$$

$$\delta(q_4, \epsilon, z_0) = (q_5, z_0)$$

State	Input	Stack Top	New Stock	Next Stage
q_0	0	z_0	az_0	q_1
q_1	0	a	aa	q_1
q_1	1	a	a	q_2
q_2	1	a	a	q_2
q_2	1	z_0	bz_0	q_2
q_2	1	b	bb	q_2
q_2	2	b	ϵ	q_3
q_3	2	b	ϵ	q_3
q_3	3	a	ϵ	q_4
q_4	3	a	ϵ	q_4
q_4	ϵ	az_0	z_0	q_f

Q3. Construct Turing machine using tape diagram, transition diagram, transition table, write transitions for the following:

1.) $\{0^n 1^n 2^n \mid n \geq 1\}$
 $L = \{001122, \dots\}$



$$S(q_0, 0) = S(q_1, A, R)$$

$$S(q_1, 0) = S(q_1, 0, R)$$

$$S(q_1, 1) = S(q_2, B, R)$$

$$S(q_2, 1) = S(q_2, 1, R)$$

$$S(q_2, 2) = S(q_3, C, R)$$

$$S(q_3, 2) = S(q_2, 2, L)$$

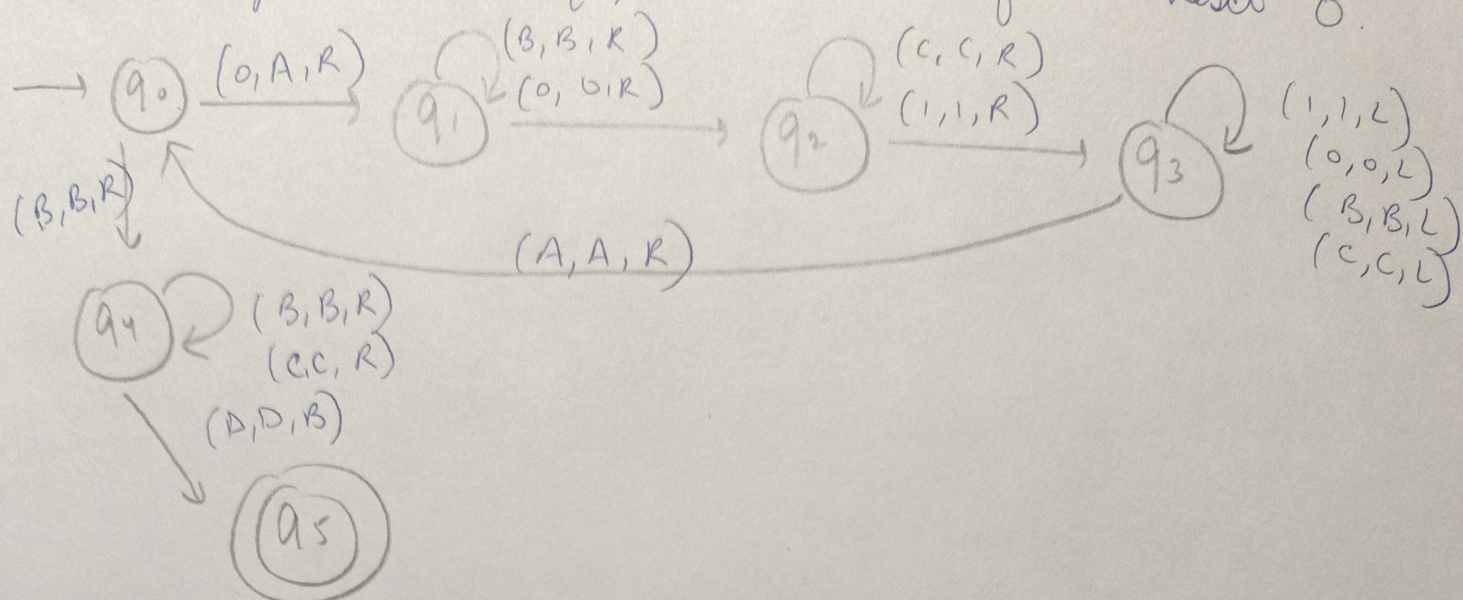
11th move $S(q_3, C) = S(q_3, C, L)$

$$S(q_3, B) = S(q_3, B, L)$$

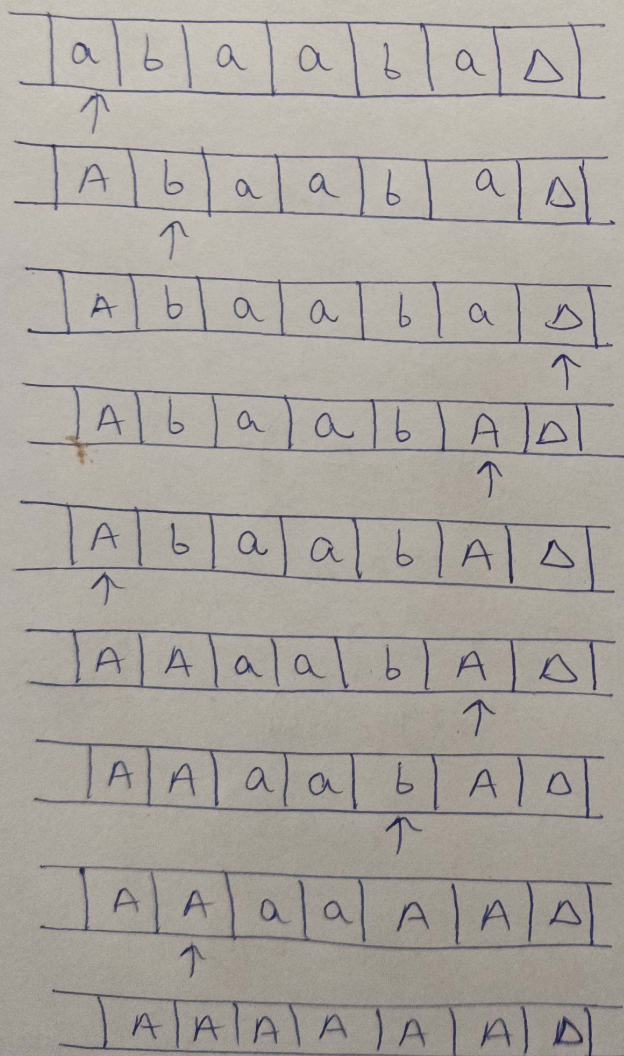
$$S(q_3, 0) = S(q_3, 0, L)$$

$$S(q_3, A) = S(q_0, A, R)$$

means go back to q_0 , and start from next 0.



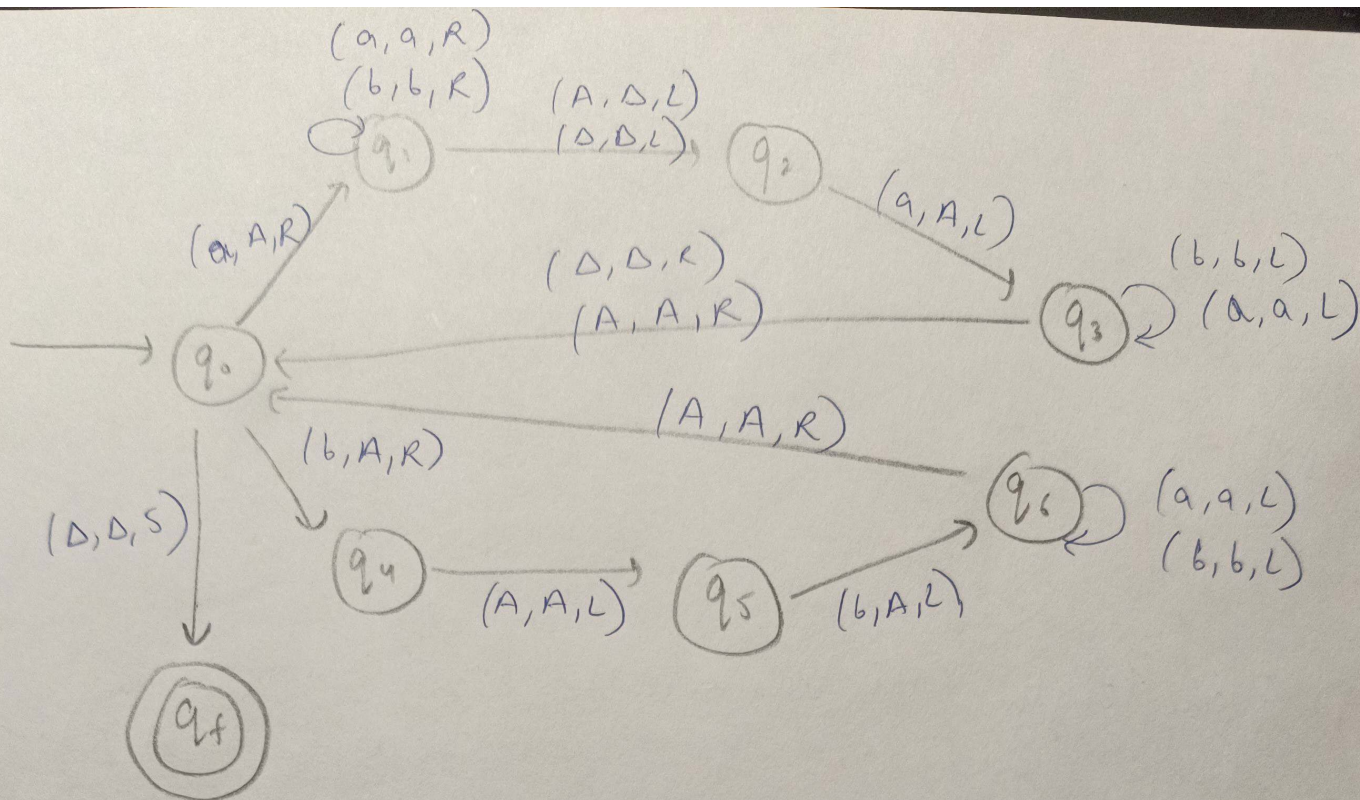
2) Even length Palindrome
 $L = \{abaaba, \dots\}$



$$\begin{aligned}
 S(q_0, a) &= S(q_1, A, R) \\
 S(q_1, a) &= S(q_1, a, R) \\
 S(q_1, b) &= S(q_1, b, R) \quad \left. \begin{array}{l} \text{Until} \\ \text{we} \\ \text{reach } \Delta \end{array} \right\} \\
 S(q_1, \Delta) &= S(q_2, \Delta, L) \\
 S(q_2, a) &= S(q_3, A, L) \\
 S(q_3, a) &= S(q_3, a, L) \\
 S(q_3, b) &= S(q_3, b, L) \quad \left. \begin{array}{l} \text{Until we} \\ \text{reach} \\ A \end{array} \right\} \\
 S(q_3, A) &= S(q_0, A, R) \\
 S(q_0, b) &= S(q_4, A, R) \\
 S(q_4, a) &= S(q_4, a, R) \\
 S(q_4, b) &= S(q_4, b, R) \quad \left. \begin{array}{l} \text{Until we} \\ \text{reach} \\ A \end{array} \right\} \\
 S(q_4, A) &= S(q_5, A, L) \\
 S(q_5, b) &= S(q_6, A, L) \\
 S(q_6, a) &= S(q_6, a, L) \\
 S(q_6, b) &= S(q_6, b, L) \quad \left. \begin{array}{l} \text{Until} \\ \text{reach} \\ A \end{array} \right\} \\
 S(q_6, A) &= S(q_0, A, R)
 \end{aligned}$$

||ly continue the process

whether every element is A. $S(q_0, A) \rightarrow$ traverse full tape and check
 $\rightarrow S(q_0, A, S)$. If yes then end the process.



3) Odd length palindrome
 $L = \{ababa, \dots\}$

a	b	a	b	a	Δ
---	---	---	---	---	---

↑

A	b	a	b	a	Δ
---	---	---	---	---	---

↑

A	b	a	b	a	Δ
---	---	---	---	---	---

↑

A	b	a	b	a	Δ
---	---	---	---	---	---

↑

A	b	a	b	A	Δ
---	---	---	---	---	---

↑

A	b	a	b	A	Δ
---	---	---	---	---	---

↑

A	b	a	b	A	Δ
---	---	---	---	---	---

↑

A	A	a	b	A	Δ
---	---	---	---	---	---

↑

A	A	a	b	A	Δ
---	---	---	---	---	---

↑

$$S(q_0, a) = S(q_1, A, R)$$

$$S(q_1, a) = S(q_1, a, R) \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{until reach } \Delta$$

$$S(q_1, b) = S(q_1, b, R)$$

$$S(q_1, \Delta) = S(q_2, \Delta, L)$$

$$S(q_2, a) = S(q_3, A, L)$$

$$S(q_3, a) = S(q_3, a, L) \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{until we reach A}$$

$$S(q_3, b) = S(q_3, b, L)$$

$$S(q_3, A) = S(q_0, A, R)$$

$$S(q_0, b) = S(q_4, A, R)$$

$$S(q_4, a) = S(q_4, a, R) \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{until we reach A}$$

$$S(q_4, b) = S(q_4, b, R)$$

$$S(q_4, A) = S(q_5, A, L)$$

A	A	a	b	A	Δ
---	---	---	---	---	---



A	A	a	A	A	Δ
---	---	---	---	---	---



A	A	a	A	A	Δ
---	---	---	---	---	---



A	A	A	A	A	Δ
---	---	---	---	---	---



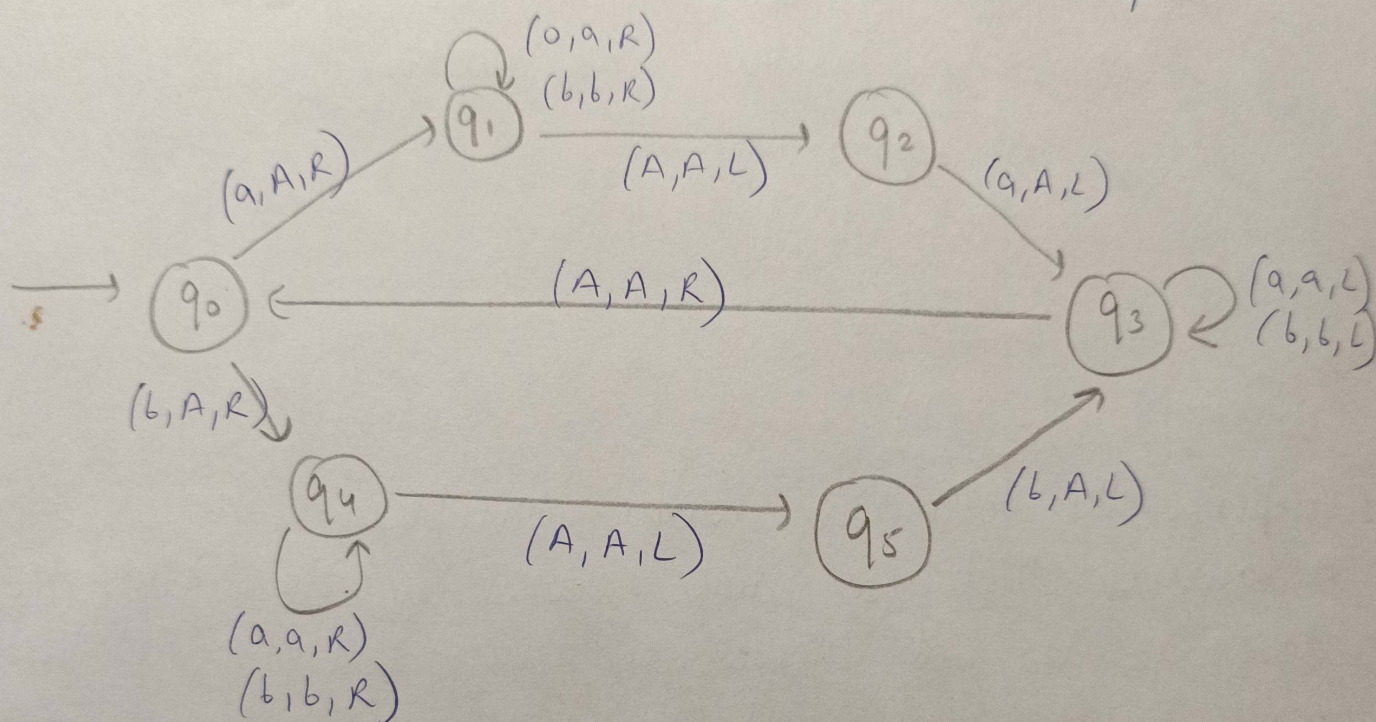
$$S(q_5, b) = S(q_3, A, L)$$

$$S(q_3, a) = S(q_3, a, L)$$

$$S(q_3, A) = S(q_0, A, R)$$

$$S(q_0, a) = S(q_1, A, R)$$

⇒ Traverse the full tape until we reach Δ then stop.



Q4. Explain the different Complexity classes with examples?

Ans There exists some problems whose solutions are not yet found, the problems are divided into classes known as Complexity classes.

Types of Complexity classes.

1) P Class (Polynomial time)

The problems in this class can be solved in polynomial time. eg: merge sort - $O(n \log n)$.

2) NP - Class

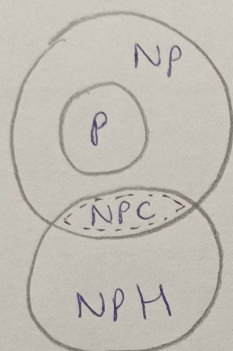
The NP Class stands for non polynomial time. Here the problems cannot be solved in polynomial time but can be verified using polynomial time. eg: Graph Coloring.

3) NP Hard Class

An NP Hard problem is at least as hard as the hardest problem in NP and it is class of problems such that every problem in NP reduces to NP-Hard.
eg: Halting problem.

4) NP Complete Class

A problem is NP Complete if it is both NP and NP-Hard. NP Complete problems are the hardest problems of NP.
eg: Travel Sales person, Non Halting problem.



Qs. Explain the following with examples -

1) A language that is not recursively enumerable (RE)

A language is not recursively enumerable (RE) if there is no Turing machine that can even semi-decide it, meaning the Turing machine cannot recognize strings belonging to the language, even if it runs forever.

example - $L = \{w \mid M_w \text{ does not halt on } w\}$, $M_w \rightarrow$ Turing machine encoded by w .

The language is not RE because -

1) $w \in L$, M_w does not halt on w , no Turing machine can determine this.

2) There is no algorithm to recognize the strings in L since it requires solving the halting problem, which is undecidable.

2) Problem that cannot be solved by Turing Machine.
A problem cannot be solved using Turing machine if it requires deciding a property that is undecidable or if

it involved a language that is not recursively enumerable.
 Example - Halting problem.

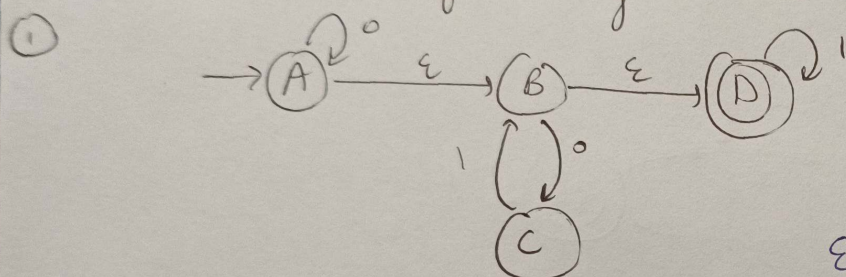
The Halting problem is undecidable because there is no general algorithm that can predict whether M halts.

3) An undecidable problem that is recursively enumerable.
 An undecidable problem that is RE means there exists a TM that can recognize instances where answer is 'yes'.
 (it halts on such inputs)

Example - Halting problem restricted to halting cases.

$$L = \{ \langle M, w \rangle \mid M \text{ halts on } w \}$$

Q6. Convert the following ϵ -NFA to NFA.



$$\epsilon \text{ closure}(C) = \{C\}$$

$$\epsilon \text{ closure}(D) = \{D\}$$

$$\epsilon \text{ closure}(A) = \{A, B, D\}$$

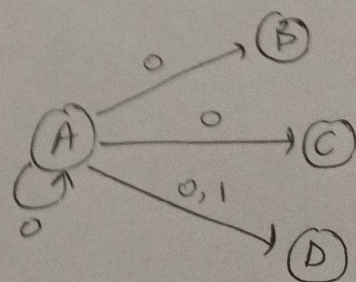
$$\epsilon \text{ closure}(B) = \{B, D\}$$

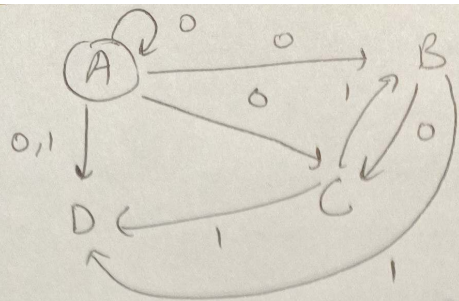
$$\begin{aligned} S'(A, 0) &= \epsilon \text{ closure}(S(\epsilon \text{ closure}(A)), 0) \\ &= \epsilon \text{ closure}(S(A, 0) \cup S(B, 0) \cup S(D, 0)) \\ &= \epsilon \text{ closure}(A \cup C \cup \phi) \\ &= \{A, B, D, C\} \end{aligned}$$

$$\begin{aligned} S'(A, 1) &= \epsilon \text{ closure}(S(\epsilon \text{ closure}(A)), 1) \\ &= \epsilon \text{ closure}(S(A, 1) \cup S(B, 1) \cup S(D, 1)) \\ &= \epsilon \text{ closure}(\phi \cup \phi \cup D) \\ &= \{D\} \end{aligned}$$

$$\begin{aligned} S'(B, 0) &= \epsilon \text{ closure}(S(\epsilon \text{ closure}(B)), 0) \\ &= \epsilon \text{ closure}(S(B, 0) \cup S(D, 0)) \\ &= \epsilon \text{ closure}(C \cup \phi) = \{C\} \end{aligned}$$

$$S'(B, 1) = \epsilon \text{ closure}(S(B, 1) \cup S(D, 1)) = \{D\}$$



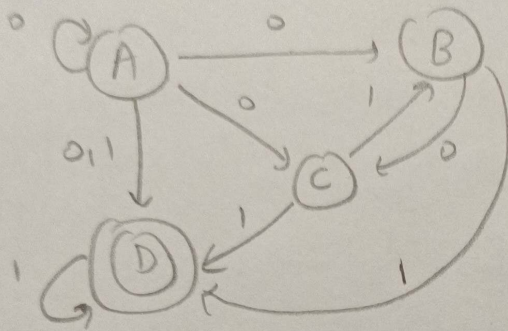


$$S'(c,0) = \epsilon\text{-closure}(S(c,0)) = \{\phi\}$$

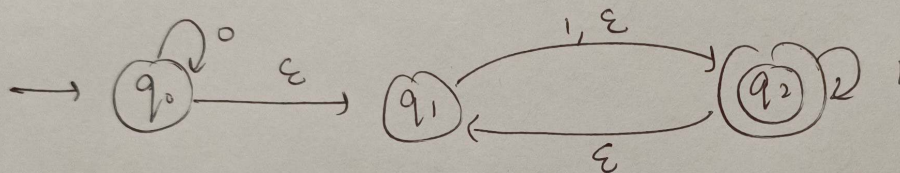
$$S'(c,1) = \epsilon\text{-closure}(S(c,1)) = \{B, D\}$$

$$S'(D,0) = \epsilon\text{-closure}(S(D,0)) = \{\phi\}$$

$$S'(A,1) = \epsilon\text{-closure}(S(A,1)) = \{D\}$$



2.)



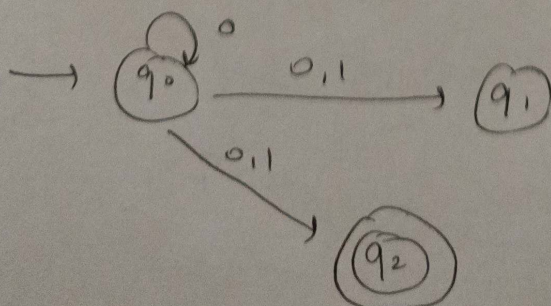
$$\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$$

$$\epsilon\text{-closure}(q_2) = \{q_2, q_1\}$$

$$\epsilon\text{-closure}(q_1) = \{q_1, q_2\}$$

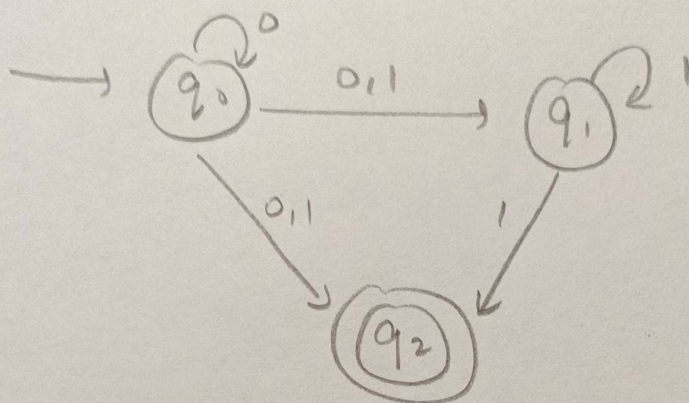
$$\begin{aligned} S'(q_0,0) &= \epsilon\text{-closure}(S(\epsilon\text{-closure}(q_0),0)) \\ &= \epsilon\text{-closure}(S(\{q_0, q_1, q_2\},0)) \\ &= \epsilon\text{-closure}(S(q_0,0) \cup S(q_1,0) \cup S(q_2,0)) \\ &= \epsilon\text{-closure}(q_0 \cup \phi \cup \phi) \\ &= \epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\} \end{aligned}$$

$$\begin{aligned} S'(q_0,1) &= \epsilon\text{-closure}(S(\epsilon\text{-closure}(q_0),1)) \\ &= \epsilon\text{-closure}(S(q_0,1) \cup S(q_1,1) \cup S(q_2,1)) \\ &= \epsilon\text{-closure}(\phi \cup q_2 \cup q_2) = \{q_1, q_2\} \end{aligned}$$



$$\begin{aligned} S'(q_1, 0) &= \varepsilon\text{-closure}(S(q_1, 0) \cup S(q_2, 0)) \\ &= \{\phi\} \end{aligned}$$

$$\begin{aligned} S'(q_1, 1) &= \varepsilon\text{-closure}(S(q_1, 1) \cup S(q_2, 1)) \\ &= \varepsilon\text{-closure}(q_2 \cup q_2) = \{q_1, q_2\} \end{aligned}$$



$$S'(q_2, 0) = \varepsilon\text{-closure}(S(q_2, 0) \cup S(q_1, 0)) = \phi$$

$$S'(q_2, 1) = \varepsilon\text{-closure}(S(q_2, 1) \cup S(q_2, 1)) = \{q_1, q_2\}$$

